Text-Conditioned Diffusion Model for Handwritten Digit Generation

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1 Introduction

Diffusion models have recently emerged as a powerful class of generative models capable of producing high-quality images through a progressive denoising process. Unlike traditional generative approaches, diffusion models learn to generate data by reversing a gradual corruption process, leading to stable training and strong generative performance.

In this project, we implement a Denoising Diffusion Probabilistic Model (DDPM) to generate handwritten digit images using the MNIST dataset. We first construct an unconditional baseline model and subsequently extend it to a conditional framework that enables controlled generation based on digit labels (0–9). The objective is to demonstrate how conditioning enhances both interpretability and structural consistency in generated outputs.

2 Method

2.1 Diffusion Process

The DDPM framework consists of two key processes:

Forward Process: A clean image x_0 is progressively corrupted by adding Gaussian noise over T timesteps:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

where $\bar{\alpha}_t$ defines the noise schedule.

Reverse Process: A neural network $\epsilon_\theta(x_t, t)$ is trained to predict the noise added at each timestep, enabling reconstruction of the original image.

The training objective is:

$$\mathcal{L} = E\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$$

2.2 Model Architecture

We employ a U-Net architecture as the backbone of the diffusion model. The architecture consists of:

- Downsampling layers to capture global contextual information
- Upsampling layers to reconstruct fine-grained spatial details
- Skip connections to preserve high-resolution features

Temporal information is incorporated through sinusoidal positional embeddings, which encode the diffusion timestep and are injected into the network.

2.3 Conditional Generation

To enable controlled image generation, we introduce label conditioning:

- Each digit label (0–9) is mapped to a learned embedding
- The label embedding is combined with the timestep embedding
- The combined embedding is injected into the U-Net layers

This conditioning mechanism allows the model to generate digit-specific outputs while preserving diversity.

3 Experiments

3.1 Dataset

We use the MNIST dataset, which contains 60,000 grayscale images of handwritten digits with a resolution of 28×28.

3.2 Training Setup

- Batch size: 64
- Number of epochs: 30
- Noise schedule: Cosine schedule
- Loss function: Mean Squared Error (MSE)

Training exhibits stable convergence, with the loss reaching approximately 0.0363 around epoch 25–30.

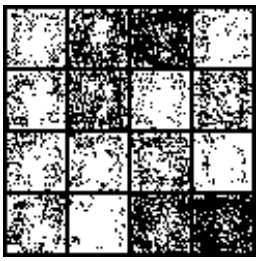
4 Results

4.1 Unconditional Model

The unconditional diffusion model generates digit-like structures without control over the output class. The generated samples exhibit moderate noise and limited structural consistency.

4.2 Conditional Model

The conditional model demonstrates significantly improved performance. Generated digits align closely with their corresponding labels, indicating successful incorporation of conditioning information.



(a) Unconditional baseline



(b) Conditional model

Figure 1: Comparison between unconditional and conditional diffusion models

5 Visualization

Figure 1 presents a comparison between the unconditional and conditional models.

Figure 2 illustrates an 8×8 grid of samples generated by the conditional model.

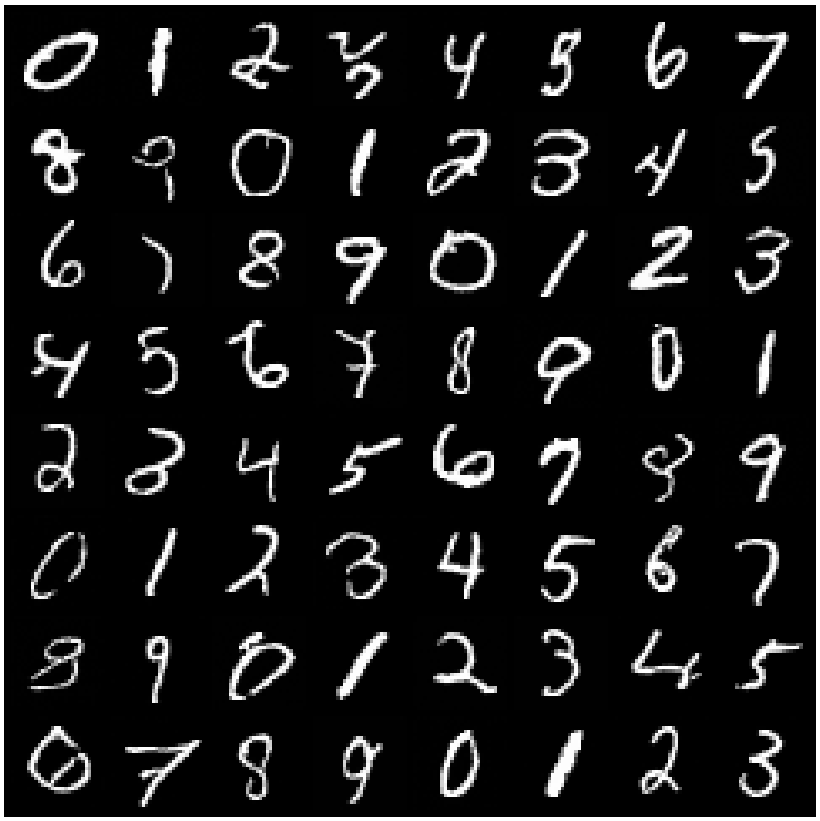


Figure 2: 8×8 grid demonstrating diversity and consistency across generated digits

The visualization highlights two key properties:

- **Consistency:** Each digit remains recognizable across multiple samples
- **Diversity:** The model produces varied stylistic representations for the same digit

6 Discussion

The conditional diffusion model achieves strong performance, producing structured and interpretable digit images aligned with input labels. Compared to the baseline, conditioning significantly enhances both control and visual coherence.

While most digits are clearly recognizable, minor ambiguity is observed in certain classes (e.g., 4 and 5). This can be attributed to the inherent variability in handwritten digit representations and the compact model architecture. Nevertheless, the model demonstrates stable convergence and effectively captures the underlying data distribution.

7 Conclusion

This project successfully implements a DDPM framework and extends it to a conditional generative model. The results demonstrate that label conditioning substantially improves control over generated outputs while maintaining diversity.

Future work may explore larger architectures, extended training, and advanced guidance techniques to further enhance image quality and generation fidelity.